AXI4 DMA Dual-Channel Subsystem

Functional Verification Report

1. Executive Summary

This report presents the RTL and UVM functional verification results for the AXI4 DMA dual-channel subsystem. The design integrates two DMA channels, two external AXI masters, a 4-to-2 AXI crossbar, a 2-to-2 AXI-Stream switch, per-channel control registers, descriptor management, global interrupt/status logic, and two 64 KiB memory targets.

The final merged regression completed on August 9, 2026 with Synopsys VCS/URG V-2023.12-SP2. All 31 maintained tests passed. Every test completed with UVM_ERROR=0 and UVM_FATAL=0, no AMD VIP protocol violations, no project SVA failures, successful explicit data comparisons, and a clean scoreboard end check.

Verification Status: PASS - Planned functional verification scope completed.

31/31	Regression Tests Passed	100%	Pass Rate	95.93%	Overall Coverage	100%	Functional Groups

Item	Result
Verification level	Integrated dual-DMA subsystem
Methodology	SystemVerilog + UVM 1.2 + UVM RAL
Simulator and coverage	Synopsys VCS/URG V-2023.12-SP2
Regression baseline	August 9, 2026 - 31/31 PASS
Source revision	Git 01b79ed (August 24, 2026)
Report date	August 25, 2026

The final overall coverage score is 95.93%: Line 99.36%, Condition 93.60%, Toggle 91.40%, FSM 100.00%, Branch 100.00%, and Functional Group 100.00%. Assertion coverage at the DUT hierarchy is 91.23%.

2. Design Under Verification and Configuration

The verification target is centered on axi_dma_subsystem_core. The synthesizable RAM wrapper provides two memory targets, while the AMD VIP testbench replaces those targets with MEM0 and MEM1 slave VIPs to provide controlled latency, backpressure, response-error injection, and passive observation.

2.1 Fixed Configuration

Configuration	Value	Verification relevance
DMA channels	2 (DMA0, DMA1)	Dual-channel concurrency, reverse completion order, and route contention
External AXI masters	2 (EXT0, EXT1)	Four crossbar inputs when combined with the two DMA sources
AXI data/address/input ID	32 / 32 / 4 bits	Memory-side output IDs widen to 6 bits to preserve source identity
DMA maximum burst	16 beats	DMA traffic uses INCR; external masters exercise FIXED, INCR, and WRAP
AXI-Lite	32-bit address and data	Separate 4 KiB windows for CH0, CH1, and GLOBAL
AXI-Stream	32-bit DATA, 4-bit KEEP	2-to-2 switch; 2-bit ID and 1-bit DEST/USER
Descriptor length / tag	20 bits / 8 bits	Legal boundaries, counters, and hardware-tag rollover
Per-channel FIFO	32 beats	Transfer, backpressure, and cleanup paths
Feature parameters	SG=0, UNALIGNED=0	Non-SG operation with 4-byte aligned DMA transfers

2.2 Address Map

Target or register block	Base address	End address or window
RAM0	0x0000_0000	0x0000_FFFF (64 KiB)
RAM1	0x1000_0000	0x1000_FFFF (64 KiB)
DMA CH0 control	0x0000_0000	4 KiB
DMA CH1 control	0x0000_1000	4 KiB
Global interrupt/status	0x0000_2000	4 KiB

2.3 Key RTL Modules

Module	Responsibility
axi_dma_subsystem_core	Integrates the AXI-Lite crossbar, 4-to-2 AXI crossbar, two DMA wrappers, AXIS switch, and descriptor, route, and interrupt control.
dma_ctrl_regs	Implements per-channel control, status, source, destination, length, route, tag, result, counter, and version registers.
dma_desc_manager	Validates commands, allocates hardware tags, acquires routes, issues descriptors, correlates status, maps errors, and records completions.
dma_channel_wrap	Wraps the AXI DMA read/write engines, AXIS FIFO, and status channels.
dma_axis_route_ctrl	Maintains destination ownership and handles requests, contention, grants, release, and route faults.
dma_irq_status_ctrl	Stores done, error, and fault pending state; applies enable, mask, and clear operations; generates channel and global interrupts.

2.4 Descriptor-Manager State Flow

IDLE -> VALIDATE -> WAIT_ROUTE -> ISSUE_WR_DESC -> ISSUE_RD_DESC -> WAIT_STATUS -> COMPLETE -> IDLE. Error handling proceeds through ERROR_RECOVERY before returning to IDLE. Validation covers channel enable, length, alignment, address range, overlap, busy state, and route legality; completion handling covers AXI responses, tag and length consistency, and abort classification.

3. Verification Environment

The static testbench tb_axi_dma_core_amd_vip owns clock and reset generation, the DUT instance, five AMD AXI VIP instances, passive internal probes, restricted negative-test hooks, and SVA bindings on both memory-side AXI interfaces. UVM class components receive typed VIP handles through configuration objects and use public APIs and analysis ports.

3.1 AMD AXI VIP Configuration

Role	Mode	Address/Data/ID	Purpose
AXI-Lite CPU	Master	32 / 32 / 0	Register frontdoor access, WSTRB, invalid-address, and timing scenarios
EXT0	AXI4 master	32 / 32 / 4	External source 0; RAM0/RAM1 paths, burst shapes, and attributes
EXT1	AXI4 master	32 / 32 / 4	External source 1; RAM0/RAM1 paths and concurrency
MEM0	AXI4 slave	32 / 32 / 6	RAM0 behavior, latency, backpressure, and SLVERR/DECERR injection
MEM1	AXI4 slave	32 / 32 / 6	RAM1 behavior, latency, backpressure, and SLVERR/DECERR injection

3.2 UVM Components and Responsibilities

Component	Responsibility
dma_subsys_vip_manager	Creates, starts, validates, and publishes five typed VIP agents. Memory slave agents start before master agents.
memory_behavior_controller	Configures deterministic memory responses, READY behavior, B/R delay, and error injection.
axil_sequencer / driver	Converts project register transactions into public AMD AXI-Lite API calls.
virtual_sequencer	Coordinates the lower-level handles required by system sequences.
cmd / route / irq monitors	Observe command acceptance and tags, route lifecycle, completion, interrupt, fault, busy, and reset events.
vip_transaction_adapter	Converts AMD monitor transactions into project-owned register and memory transactions.
reference_model	Maintains an independent sparse, byte-addressable expected memory.
scoreboard	Correlates intent, command, route, bus traffic, completion, registers, and interrupts; checks data and lifecycle state.
coverage	Samples 11 system-domain functional covergroups.
RAL + predictor	Uses a 4-byte little-endian frontdoor map; the passive predictor updates mirrors from observed AXI-Lite traffic.
watchdog	Detects global hangs and excessive route wait time.

3.3 Self-Checking Data Path

- Virtual sequences publish explicit expected intent before starting an operation, including route, length, tags, acceptance, completion, interrupt behavior, and expected errors.
- VIPs and passive monitors record actual AXI-Lite, AXI, route, completion, interrupt, and fault activity.
- The independent reference model maintains expected source and destination bytes and commits destination data only after a successful modeled operation.
- The scoreboard decodes widened crossbar IDs to identify DMA0, DMA1, EXT0, and EXT1, then checks byte-level data, tags, lengths, errors, ordering, route release, and interrupts.
- check_phase confirms that no intent, active flow, route ownership, or data comparison remains outstanding.

4. Regression Execution and Results

With no test argument, run_vcs_core_amd_vip.sh compiles once, executes 31 unique UVM_TESTNAME values, merges coverage, and generates one URG report. Each test retains an individual simulation log and coverage identity. The final VDB is reported with the reviewed strict exclusion file.

4.1 Regression Verdict

- All 31 maintained tests returned a successful status: 31/31 PASS, 100% pass rate.
- Every test completed with UVM_ERROR=0 and UVM_FATAL=0.
- AMD VIP protocol checking, project SVA, explicit sequence comparisons, and scoreboard check_phase all passed.
- The merged coverage VDB and URG dashboard were generated successfully at 01:31:37 on August 9, 2026.

4.2 Results by Scenario Group

Scenario group	Tests	Result	Primary verification objective
Infrastructure	4	4/4 PASS	VIP bring-up, complete environment connectivity, RAL frontdoor access, and passive prediction
Nominal DMA, ordering, and routing	6	6/6 PASS	Bidirectional transfers, full path matrix, completion reordering, route contention, and release
Length, burst, and AXI attributes	6	6/6 PASS	Length boundaries, 64 KiB extent, FIXED/INCR/WRAP, attributes, and bus toggles
Registers and AXI-Lite	3	3/3 PASS	Access policy, WSTRB, AW/W independence, B/R stability, and condition closure
Errors, faults, and abort	6	6/6 PASS	Command rejection, memory response errors, status/fault protection, and abort timing
Interrupt and reset	4	4/4 PASS	Multiple pending causes, masking, status separation, active reset, and FSM reset arcs
Counter and tag rollover	2	2/2 PASS	Counter low-byte rollover and hardware-tag transitions on legal traffic
Total	31	31/31 PASS	All planned regression scenarios completed successfully

4.3 Key Closure Scenarios

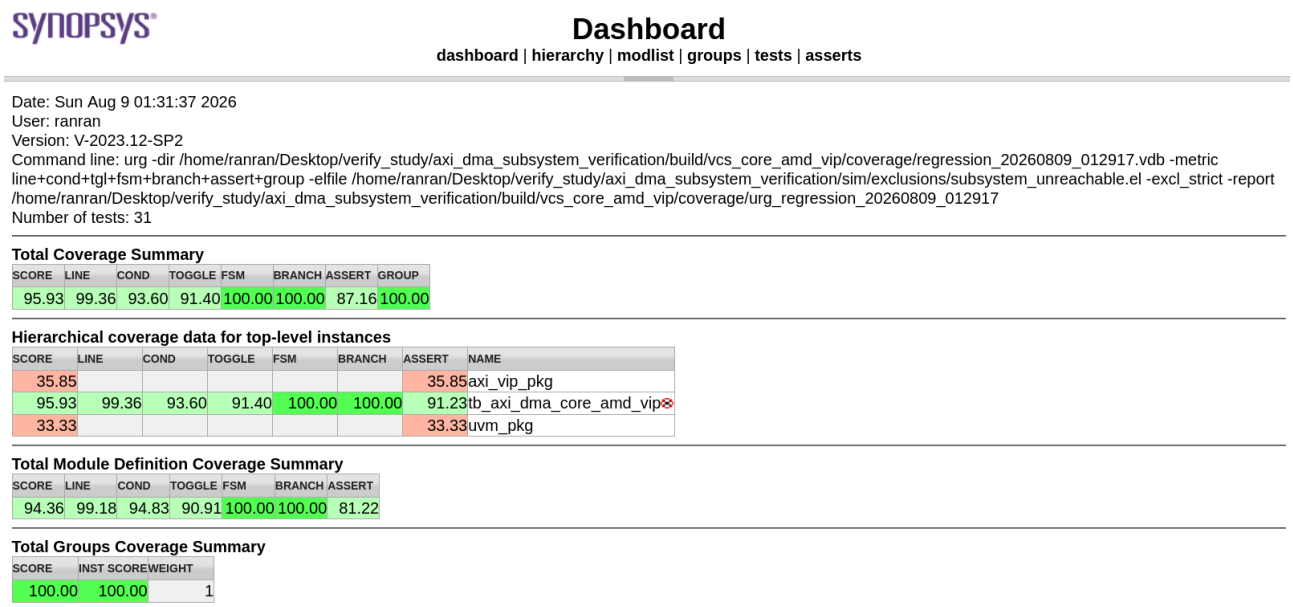
- Bidirectional DMA transfers and every legal DMA0/DMA1/EXT0/EXT1 to RAM0/RAM1 crossbar path.
- Both issue/completion ordering combinations, route contention, wait accounting, ownership, and release.
- Zero-length rejection, all software-visible length classes, and legal single-window transfers from 4 bytes through 65,536 bytes.
- External-master FIXED, INCR, and WRAP bursts, including legal 256-beat INCR traffic.
- AXI SLVERR/DECERR, descriptor-validation errors, tag/length/status faults, interrupt masking, and multiple pending causes.
- Idle and active reset, every reachable nonterminal FSM reset arc, pending and in-flight abort behavior, and post-reset reuse.

5. Coverage Closure

The final URG dashboard was generated from the merged 31-test VDB with sim/exclusions/subsystem_unreachable.el applied in strict mode. Structural coverage is scoped to tb_axi_dma_core_amd_vip.dut; functional covergroups and project assertions remain independently reported.

5.1 Final Coverage Summary

Metric	Final result	Closure interpretation
URG overall score	95.93%	Merged result from all 31 tests
Line	99.36%	Reachable RTL statements exercised
Condition	93.60%	Residual objects audited for configuration and temporal reachability
Toggle	91.40%	Programmable addresses, data, attributes, tags, and buses exercised
FSM	100.00%	All reachable states and legal transitions covered
Branch	100.00%	All reachable branches covered
Assertion - aggregate report	87.16%	Includes instrumented axi_vip_pkg and uvm_pkg statistics
Assertion - DUT hierarchy	91.23%	Result at tb_axi_dma_core_amd_vip.dut
Functional group	100.00%	All 11 planned functional covergroups closed



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Dashboard

dashboard | hierarchy | modlist | groups | tests | asserts

Date: Sun Aug 9 01:31:37 2026
User: ranran
Version: V-2023.12-SP2
Command line: urg -dir /home/ranran/Desktop/verify_study/axi_dma_subsystem_verification/build/vcs_core_amd_vip/coverage/regression_20260809_012917.vdb -metric line+cond+tgl+fsm+branch+assert+group -elfile /home/ranran/Desktop/verify_study/axi_dma_subsystem_verification/sim/exclusions/subsystem_unreachable.el -excl_strict -report /home/ranran/Desktop/verify_study/axi_dma_subsystem_verification/build/vcs_core_amd_vip/coverage/urg_regression_20260809_012917
Number of tests: 31

Total Coverage Summary

SCORE	LINE	COND	TOGGLE	FSM	BRANCH	ASSERT	GROUP
95.93	99.36	93.60	91.40	100.00	100.00	87.16	100.00

Hierarchical coverage data for top-level instances

SCORE	LINE	COND	TOGGLE	FSM	BRANCH	ASSERT	NAME
35.85						35.85	axi_vip_pkg
95.93	99.36	93.60	91.40	100.00	100.00	91.23	tb_axi_dma_core_amd_vip
33.33						33.33	uvm_pkg

Total Module Definition Coverage Summary

SCORE	LINE	COND	TOGGLE	FSM	BRANCH	ASSERT
94.36	99.18	94.83	90.91	100.00	100.00	81.22

Total Groups Coverage Summary

SCORE	INST SCORE	WEIGHT
100.00	100.00	1

Figure 1. Final Synopsys URG dashboard - 31-test merged regression, August 9, 2026

6. Functional Coverage Model

dma_subsys_coverage defines 11 system-level covergroups with 65 coverpoints and 8 crosses. Final Functional Group coverage is 100.00%; command, register, memory-protocol, routing, completion, interrupt, fault, and reset objectives are all covered.

Covergroup	Coverage dimensions	Result
command_cg	Accepted source and destination channels, length classes, and route cross	100%
intent_cg	Intended route and length, result class, and expected acceptance, completion, and interrupt behavior	100%
register_cg	Register block and address, access type, WSTRB, response, and access-response cross	100%
memory_cg	DMA0/DMA1/EXT0/EXT1 origin, RAM0/RAM1 target, access, response, and path crosses	100%
memory_protocol_cg	Burst, beat count, size, ID, alignment, 4 KiB rule, strobe class, and malformed traffic	100%
route_cg	Request, grant, release, fault, source, destination, wait class, and path cross	100%
completion_cg	Owner, error/result class, abort state, and owner-abort cross	100%
completion_order_cg	Issue order, completion order, reordering, and pending depth	100%
irq_cg	Pending and enable state, channel/global interrupt, causes, masking, and consistency	100%
fault_cg	Manager/route fault type, source, and enable state	100%
reset_cg	Assertion and deassertion, boot/repeated epoch, and idle/active context	100%

6.1 Coverage-to-Test Mapping

Coverage objective	Principal tests
Environment and VIP bring-up	VIP smoke, VIP manager smoke, and environment smoke
RAL and register prediction	RAL smoke and register access policy
Data movement and path matrix	CH0-to-CH1, CH1-to-CH0, and crossbar path matrix
Completion order and route arbitration	Completion reorder/reverse, route contention, and abort timing
Length, burst, and attributes	Transfer-length boundary, legal extent, AXI burst shape, and attribute toggle
Error, fault, and interrupt	Descriptor error, memory response error, status fault, and interrupt multi-pending/separation
Reset, FSM, and rollover	Reset recovery, FSM reset arc, tag-counter rollover, and legal-tag rollover

7. Assertions and Protocol Checking

The project checker dma_subsys_axi_sva defines 21 assertions and 5 cover properties and is bound to both memory-side AXI interfaces. AMD AXI VIP provides concurrent vendor protocol checking, creating two complementary protocol-verification layers.

Check family	Checked behavior	Activation scenario
Reset clearing	AW, W, and AR request VALID signals clear after reset	Reset recovery and FSM reset arc
Payload stability	AW, W, AR, B, and R payloads remain stable while VALID is asserted and READY is low	AXI backpressure and memory delay/error
Known-value checks	Addresses, controls, data, responses, and IDs contain no X/Z values on valid channels	Continuous throughout the regression
Size and burst legality	AWSIZE and ARSIZE are legal; reserved burst encodings are rejected	Burst shape and attribute toggle
FIXED constraints	FIXED burst length does not exceed the configured limit	AXI burst shape
WRAP constraints	WRAP length is one of 2, 4, 8, or 16 beats	AXI burst shape
4 KiB boundary	INCR bursts do not cross the AXI 4 KiB boundary	Path, boundary, and burst scenarios
Cover properties	AW/W/AR backpressure plus FIXED and WRAP activation	Directed protocol-activation tests

7.1 Protocol Verdict

- No project assertion failures occurred in the 31-test regression.
- AMD VIP reported no AXI or AXI-Lite protocol violations.
- DUT-hierarchy assertion coverage is 91.23%; the aggregate assertion metric is 87.16%.
- Directed scenarios activated AW/W/AR backpressure and the FIXED and WRAP cover properties.

8. Scoreboard, Reference Model, and RAL

The environment cross-checks sequence intent, actual observed activity, and an independent reference model. The checker does not reuse the DUT burst splitter, FIFO pointers, or descriptor-manager state machine.

8.1 Scoreboard Checks

- Command fields: source, destination, route, length, software tag, and hardware tag.
- Data path: crossbar-ID ownership, read/write addresses, byte strobes, byte-level data, and destination memory after completion.
- Lifecycle: route request, grant, wait, release, concurrent pending depth, and issue/completion order.
- Completion and status: owner, hardware/software tags, error, completed length, abort class, and register-visible state.
- Interrupt and fault behavior: done/error/fault pending state, masks, channel/global interrupts, and clear operations.

8.2 RAL Result

dma_subsys_reg_block contains CH0, CH1, and GLOBAL sub-blocks in a single 4-byte little-endian frontdoor map. Automatic prediction is disabled; uvm_reg_predictor updates mirrors from observed AXI-Lite transactions. RAL smoke, register-policy, AXI-Lite timing, and condition-closure tests verified RW/RO/WO behavior, WSTRB, invalid-address responses, reset values, VERSION=0x0001_0000, and mirror consistency.

9. Structural Coverage Exclusion Audit

The final exclusion file contains only structural coverage objects proven unreachable by fixed configuration or temporal reachability analysis. URG applies the file with -excl_strict, binding checksums and object IDs to the current VDB. The file does not exclude assertions or functional covergroups.

Metric or object	Directives	Audit conclusion
Line (Block)	6	Unreachable statements in the integrated command and abort handshake
Condition	22	Impossible Boolean vectors created by prerequisites, compound predicates, or direct-handshake relationships
Toggle	170	Constant, statically assigned, alignment, width, address-window, and protocol-configuration bits; one directive may cover a bit range
FSM transition	2	One impossible ISSUE_RD_DESC -> COMPLETE transition in each channel
Branch	2	Unreachable pending-only abort path and write_valid case-default path
Assertion / functional	0	Project assertions and all 11 functional covergroups remain unexcluded

9.1 Principal Toggle Exclusion Families

Object family	Fixed reason
AXI-Lite address and response bits	32-bit word alignment; local targets produce only OKAY or SLVERR.
RAM address upper bits	Each memory target is a fixed 64 KiB window; address bit 28 selects RAM1.
DMA AXI ID, size, burst, cache, and protection	Channel identity and integration parameters fix these values; corresponding programmable external-master fields are covered.
Descriptor/status USER and unused ID bits	USER is tied low and unused ID bits remain constant in this configuration.
Completion-length upper bits	A legal single-window transfer is at most 65,536 bytes; length[19:17] is unreachable.
Route-fault code and source	The route controller emits a fixed route-conflict code and fixed route source.

9.2 Key FSM, Line, and Branch Evidence

- rd_status_accept depends on registered rd_desc_issued_reg, so the legal path must pass through WAIT_STATUS; ISSUE_RD_DESC -> COMPLETE cannot occur directly.
- ST_VALIDATE lasts one clock, so a second software abort cannot traverse the AXI-Lite path and commit within the same integrated timing window.
- pending_cmd_accept is directly related to cmd_valid and cmd_ready; the pending-only abort branch cannot be entered in the integrated handshake.
- write_valid is already restricted by address selection. Invalid local offsets return SLVERR externally and do not enter the internal case-default branch.

10. AXI-Lite DECERR Fix Verification

The imported axil_crossbar_wr.v implementation was corrected so an unmapped write returns local DECERR/BVALID only after the matching W-channel handshake. An error-only W-credit counter preserves ordering for multiple outstanding writes.

Fix area	Verified result
DECERR BVALID timing	The local DECERR response is issued only after the W beat is accepted
WREADY and W-state gating	State updates occur only for a valid write command
Local/downstream response isolation	All downstream BREADY signals are suppressed while a local DECERR is at the response head
Reset consistency	w_select_reg, w_drop_reg, and w_select_valid_reg reset together
Directed verification	Register-access policy, AXI-Lite protocol timing, and condition-closure tests all passed

11. Final Conclusion

The AXI4 DMA dual-channel subsystem has completed its planned functional verification scope. The 31-test UVM regression passed in full and covered the control plane, descriptor admission, dual-channel data movement, crossbar and AXIS routing, concurrent completion ordering, AXI protocol stress, errors, faults, abort behavior, interrupts, reset recovery, boundaries, and rollover.

Sign-off item	Result
Regression	31/31 PASS - 100% pass rate
UVM / VIP / SVA	UVM_ERROR=0, UVM_FATAL=0; VIP protocol checks passed; no project SVA failures
Scoreboard and data integrity	No unmatched intent, active flow, route leak, or byte mismatch
Functional coverage	11/11 covergroups - 100.00%
Structural coverage	Line 99.36%, Condition 93.60%, Toggle 91.40%, FSM 100%, Branch 100%
Overall coverage	95.93%
Verification conclusion	PASS - Functional verification sign-off baseline for the current RTL and UVM environment

FINAL VERIFICATION STATUS: PASS

The reported results are traceable to the final URG dashboard dated August 9, 2026, the maintained regression list, the coverage model, SVA, scoreboard, RAL model, and strict exclusion file. The report source baseline is Git revision 01b79ed.

Appendix A. Complete 31-Test Regression Inventory

This inventory matches the regression_tests array in run_vcs_core_amd_vip.sh and the maintained test-scenario document. Every listed test passed.

Group	Test name	Verification objective
Infrastructure	amd_axi_vip_smoke_test	Starts and exercises all five AMD VIPs, including register reads and EXT0/EXT1 memory write/readback.
Infrastructure	dma_subsys_vip_manager_smoke_test	Checks VIP-manager configuration, handles, startup order, and lifecycle.
Infrastructure	dma_subsys_env_smoke_test	Checks complete reusable UVM environment construction, connectivity, and idle-state behavior.
Infrastructure	dma_subsys_ral_smoke_test	Checks RAL frontdoor access, passive prediction, mirrors, reset values, and version values.
Nominal/routing	dma_subsys_ch0_to_ch1_test	Checks the primary CH0-driven RAM0-to-RAM1 transfer path.
Nominal/routing	dma_subsys_ch1_to_ch0_test	Checks the reverse CH1-driven RAM1-to-RAM0 path.
Nominal/routing	dma_subsys_crossbar_path_matrix_test	Exercises every legal DMA0/1 and EXT0/1 path to RAM0/1.
Nominal/routing	dma_subsys_completion_reorder_test	Confirms correct correlation when two DMA channels complete out of issue order.
Nominal/routing	dma_subsys_completion_order_reverse_test	Exercises the opposite issue/completion ordering combination.
Nominal/routing	dma_subsys_route_contention_test	Checks destination ownership, arbitration, wait accounting, and release.
Length/AXI	dma_subsys_transfer_length_boundary_test	Exercises software-visible length classes and boundary transitions.
Length/AXI	dma_subsys_axi_burst_shape_test	Exercises legal FIXED, INCR, and WRAP burst shapes independently of DMA-generated traffic.
Length/AXI	dma_subsys_toggle_value_sweep_test	Expands structural toggling across tags, lengths, addresses, and response encodings.
Length/AXI	dma_subsys_legal_extent_test	Checks the maximum legal 65,536-byte transfer within one RAM window.
Length/AXI	dma_subsys_axi_attribute_toggle_test	Exercises programmable external-master AXI IDs, CACHE, PROT, and related attributes.
Length/AXI	dma_subsys_bus_toggle_closure_test	Closes remaining reachable bus, address, and length toggles.

Appendix A. Regression Inventory (continued)

Tests 17-31 are listed below. Every test passed.

Group	Test name	Verification objective
AXI-Lite	dma_subsys_register_access_policy_test	Checks RW/RO/WO policy, WSTRB behavior, and invalid-address responses.
AXI-Lite	dma_subsys_axil_protocol_timing_test	Checks AW/W independence and B/R stability under backpressure.
AXI-Lite	dma_subsys_axil_condition_closure_test	Hits remaining legal channel/global register and AXI-Lite integration conditions.
Error/abort	dma_subsys_descriptor_error_test	Checks disabled, zero-length, alignment, range, overlap, busy, and route-related command rejection.
Error/abort	dma_subsys_memory_response_error_test	Checks AXI read/write SLVERR and DECERR mapping into DMA completion status.
Error/abort	dma_subsys_status_fault_injection_test	Checks tag/length/status protection and manager/route fault reporting.
Error/abort	dma_subsys_status_fault_expansion_test	Adds channel, source, and encoding permutations for status faults.
Error/abort	dma_subsys_abort_timing_test	Distinguishes pending-command and in-flight abort behavior.
Error/abort	dma_subsys_abort_phase_matrix_test	Covers opposite channel roles, multiple abort phases, and subsequent recovery.
IRQ/reset	dma_subsys_irq_mask_multi_pending_test	Checks separation of pending state, enable masks, channel interrupts, and global interrupt.
IRQ/reset	dma_subsys_irq_status_separation_test	Checks independent done/error/fault storage and byte-strobe-aware clearing.
IRQ/reset	dma_subsys_reset_recovery_test	Checks idle and active reset, route and interrupt cleanup, and post-reset transfer reuse.
IRQ/reset	dma_subsys_fsm_reset_arc_test	Covers reset exits from every reachable nonterminal state and legal stalled-descriptor abort branches.
Rollover	dma_subsys_tag_counter_rollover_test	Uses 129 deterministic error commands per channel to exercise counter low-byte and tag-sequence transitions.
Rollover	dma_subsys_legal_tag_rollover_test	Uses 65 legal 4-byte transfers per channel to exercise hardware-tag bits on descriptor/status traffic.

Appendix B. Report Basis and Traceability

Source	Use in this report
axi_dma_subsystem_verification/README.md	Architecture, environment, coverage baseline, regression flow, and sign-off interpretation
docs/AXI_DMA_Subsystem_Test_Scenarios.md	Purpose, stimulus, and checks for all 31 tests
docs/AXI_DMA_Subsystem_Coverage_Waiver_Rationale.md	Reviewed Line, Condition, Toggle, FSM, and Branch exclusions and reachability evidence
docs/Coverage_report.png	Final 31-test merged URG dashboard and coverage results
docs/RTL_fix_intro.md	AXI-Lite unmapped-write DECERR/W-handshake correction
docs/AMD_axi_vip_external_dependency.md	Modes, widths, and static dependencies for the five VIP instances
rtl/dma_subsystem_pkg.sv	Channel counts, widths, memory/register addresses, error codes, and fixed parameters
rtl/axi_dma_subsystem_core.sv	Integration topology, crossbar/switch connectivity, and module instances
tb/uvm/env/dma_subsys_coverage.svh	11 covergroups, 65 coverpoints, and 8 crosses
tb/uvm/env/dma_subsys_scoreboard.svh	Intent, bus, route, completion, interrupt, data, and check_phase checks
tb/sva/dma_subsys_axi_sva.sv	21 assertions and 5 cover properties
tb/uvm/ral/dma_subsys_reg_model.svh	CH0, CH1, and GLOBAL RAL blocks, access policy, and mapping
sim/run_vcs_core_amd_vip.sh	Maintained test list, compile/regression/merge/URG flow, and PASS criteria
sim/cm_hier.cfg	Structural coverage scope at the DUT hierarchy
sim/exclusions/subsystem_unreachable.el	Strict, reviewed exclusions for unreachable structural objects